

Project 3: Ship tracking from bearings

Please send your solutions to andrea.riebler@ntnu.no.

Deadline: Thursday April, 30th, 18:00.

General comments:

- This project is compulsory. To be able to take the exam, a reasonable attempt must be made to solve all problems.
- The project report should be formulated as a scientific report. In particular, it should be possible to understand what you have done without reading the questions in this problem text. Moreover, the text in the project report should consist of full sentences and proper punctuation should be used throughout, also in equations! All results you present should be discussed. What can you (and the world) learn from your results? The project text should be written so that it is easy to follow by your fellow students in MA8702. The project report should consist of one (and only one) pdf-file. It should include derivation of formulas that you are using in your implementations. The project report should also include the code you have used to solve the project and the plots you have generated. Associated to the various plots there should be captions explaining the contents of the plots, and in addition all the plots should be referenced, explained and discussed in the main text of the report

The report can be written in English or Norwegian. The project can be done alone or in groups of two or three persons. If you do the project in a group, only one of the individuals in the group should hand in the solution via email to andrea.riebler@ntnu.no. In your solution, specify your (full) names, NOT student or candidate numbers.

Consider three different filtering approaches (Extended Kalman filter, Particle filter, Ensemble Kalman filter) for the task of tracking a ship on the surface using two bearings-only observations.

There are two sensors measuring angles to a vessel. One sensor is located in east,north coordinate $(0,0)$, the other at coordinate $(40,40)$ in km units. See Figure 1 for a map view of this situation with the initial observation and an example of a realized ship trajectory.

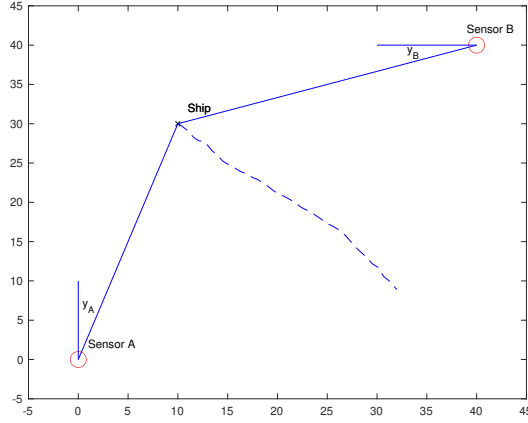


Figure 1: Sensor A and B in bearing-only tracking.

The state space model is defined as follows:

A surface vessel has state $\mathbf{x}_t = (E_t, N_t, v_t, u_t)^T$, where (E_t, N_t) is the east and north position at time t , while (v_t, u_t) is the associated velocity vector at time t .

The prior initial state is, $\mathbf{x}_1 \sim N_4(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$, where the expectation is $\boldsymbol{\mu}_1 = (10, 30, 10, -10)^T$ and the covariance matrix is $\boldsymbol{\Sigma}_1 = \text{Diag}(10^2, 10^2, 5^2, 5^2)^T$.

The prior dynamic process model for the vessel is defined by

$$\begin{aligned} \mathbf{x}_{t+1} &= \mathbf{A}\mathbf{x}_t + \boldsymbol{\epsilon}_{t+1}, \\ \mathbf{A} &= \begin{bmatrix} 1 & 0 & \delta & 0 \\ 0 & 1 & 0 & \delta \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ \boldsymbol{\epsilon}_{t+1} &\sim N_4(\mathbf{0}, \text{Diag}[0.1^2, 0.1^2, 0.5^2, 0.5^2]) \end{aligned}$$

for $t = 1, \dots, T-1$, and with $\delta = 1/60$ which means sampling interval of every minute.

The observations made at sensors A and B, $\mathbf{y}_t = (y_{A,t}, y_{B,t})^T$, $t = 1, \dots, T$, are modelled as conditionally independent and single-site response with additive

Gaussian errors:

$$\mathbf{y}_t = \begin{bmatrix} \arctan(E_t/N_t) \\ \arctan[(40 - N_t)/(40 - E_t)] \end{bmatrix} + \boldsymbol{\epsilon}_t$$

$$\boldsymbol{\epsilon}_t \sim N_2(\mathbf{0}, \text{Diag}[0.1^2, 0.1^2]),$$

where the inverse tangent function is defined as indicated in Figure 1, assuming that the ship is entering the field of interest for both sensors.

The angle observations at sensor A and B can be downloaded from <https://folk.ntnu.no/joeid/MA8702/sensorA.txt> and <https://folk.ntnu.no/joeid/MA8702/sensorB.txt>. These observations are displayed over $T = 50$ time steps in Figure 2.

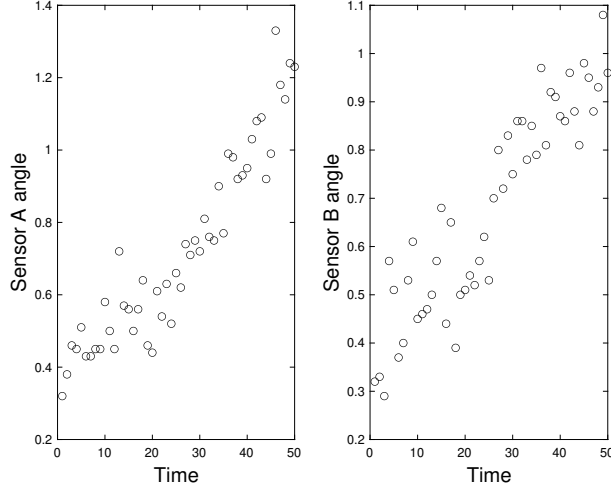


Figure 2: Observations of Sensor A and B.

TASKS

The objective of the study is to assess the filtering pdf $p(\mathbf{x}_t|\mathbf{y}_1, \dots, \mathbf{y}_t)$, $t = 1, \dots, T$.

- Implement an extended Kalman filter algorithm. This means linearizing the measurement equation around the predicted state. Plot the filtering solution over time in a map view, along with uncertainty bounds. Also indicate the position of the sensors in the map. Plot the filter variances of the position coordinates over time. Show your derivation and discuss the results.
- Implement a standard particle filter algorithm with $B = 10000$ particles, using the state Markovian process model as proposal at each time step. Apply resampling in the particle filter. Plot the filtering solution over

time in a map view, along with uncertainty bounds. Also indicate the position of the sensors in the map. Plot the filter variances of the position coordinates over time. Activate also the particle filter with only $B = 100$ particles. Discuss the results.

- Implement a standard ensemble Kalman filter with $B = 1000$ ensemble members. Plot the filtering solution over time in a map view, along with uncertainty bounds. Also indicate the position of the sensors in the map. Plot the filter variances of the position coordinates over time. Activate also the ensemble Kalman filter with only $B = 100$ ensembles. Discuss the results.

Finally, summarize your major experiences with the study.